BankOps

Banking Operations Request Management Platform

Software Requirements and System Documentation

Technologies: Power Apps · Dataverse ·Power Automate · Power BI

Document Date: 5 September 2026

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Table of Contents

1. Project Overview *4*

1.1 About BankOps **4**

1.2 Project Objectives **4**

1.3 Project Nature **4**

2. Project Scope *4*

2.1 Included Features **4**

2.2 Out of Scope **5**

3. User Roles and Access *5*

3.1 Employee **6**

3.2 Manager **6**

3.3 Admin **6**

4. Application Functions *6*

4.1 Login **6**

4.2 Home and Power Apps Dashboard **6**

4.3 Create Request **6**

4.4 My Requests **7**

4.5 Assigned Requests **7**

4.6 Request Details **7**

4.7 Request Updates and Status Management **7**

4.8 Image Management **7**

4.9 User Management **7**

4.10 Email Notifications **7**

4.11 Full Dashboard **8**

5. Functional Requirements *8*

6. Non-Functional Requirements *9*

7. Technology Stack and Architecture *10*

7.1 Technologies Used **10**

7.2 High-Level Architecture **10**

7.3 Data Model **10**

7.4 Request Lifecycle **11**

7.5 Image Storage and Removal **11**

8. External Interfaces *12*
8.1 User Interface **12**
8.2 Dataverse Interface **12**
8.3 Email Interface **12**
8.4 Reporting Interface **12**
8.5 Operating Requirements **12**
9. Use Case Overview *12*
9.1 Actors **12**
9.2 Main Use Cases **13**
10. Verification and Demonstration *13*
Appendix A — Application Screenshots *14*
Appendix B — Project Limitations and Future Improvements *19*
B.1 Project Limitations **19**
B.2 Future Improvements **19**

1. Project Overview

1.1 About BankOps

BankOps is a low-code application that simulates an internal banking operations request management system. It allows employees to submit requests, track progress, and view request details, while Managers and Admins access additional request-management and reporting functions. The application uses Microsoft Power Apps for its user interface and Microsoft Dataverse to store user records, requests, request updates, and associated images. Power Automate handles email notifications, while Power BI provides a separate, more detailed reporting dashboard.

BankOps demonstrates how business operations can be organized and monitored through a centralized digital application.

1.2 Project Objectives

- Centralize operational requests in one application.
- Allow users to create, track, and manage requests.
- Provide different navigation options and request visibility for each role.
- Maintain request details, status history, and image attachments.
- Automate assignment-related email notifications.
- Provide operational summaries through Power Apps and Power BI dashboards.
- Demonstrate practical use of Microsoft Power Platform technologies.

1.3 Project Nature

BankOps is a functional portfolio project built around a simulated banking environment. It uses fictional users and test data and is not affiliated with a real bank.

The login process validates demonstration credentials stored in Dataverse. It does not use Microsoft Entra ID authentication or provide enterprise-grade credential security. Production use would require an approved identity provider, appropriate security controls, and a review of data access permissions.

2. Project Scope

2.1 Included Features

The current application includes:

- Demo login using configured user records.
- Role-based application navigation.
- Home screen and Power Apps dashboard.

- Request creation, assignment, and tracking.
- Search and filtering of request lists.
- Request details and status management.
- Request-update history.
- Optional image attachments and image removal.
- User management for Admin.
- Assignment-related email notifications through Power Automate.
- A separate Power BI reporting dashboard.

2.2 Out of Scope

The current project does not include:

- Processing real customer, account, payment, or banking data.
- Microsoft Entra ID single sign-on or multifactor authentication.
- Production-grade password management.
- Verified regulatory or security certification for a financial institution.
- SQL or SharePoint as connected components of the current application.
- Offline operation or a standalone native mobile application.

3. User Roles and Access

BankOps provides three user roles. Each role has a dedicated navigation menu and request visibility rules.

Feature	Employee	Manager	Admin
Home	Yes	Yes	Yes
My Requests	Yes	Yes	Yes
New Request	Yes	Yes	Yes
Assigned Requests	No	Yes	Yes
Full Dashboard (Power BI)	No	Yes	Yes
User Management	No	No	Yes
View requests created by or assigned to the current user	Yes	Yes	Available alongside broader access
View all requests	No	No	Yes

3.1 Employee

Employees can create new requests, access the Home screen, and view My Requests. Their request list contains requests they created or that were assigned to them.

3.2 Manager

Managers have the standard application functions, plus Assigned Requests and Full Dashboard. Their request visibility is limited to requests they created or that were assigned to them.

3.3 Admin

Admins have access to the standard functions, Assigned Requests, Full Dashboard, and User Management. Admins can view all requests and manage configured user roles and account states.

Note: These rules describe the tested application behavior. Role-based navigation and Dataverse permissions are separate controls.

4. Application Functions

4.1 Login

Users select or enter a configured demonstration identity and submit their credentials. The application validates the login against its configured Dataverse demo-user data and opens the appropriate role-based experience.

4.2 Home and Power Apps Dashboard

The Home screen provides navigation and an overview of request activity. The internal dashboard displays request summaries, including the configured totals and status counts.

4.3 Create Request

Users enter the request subject, type, priority, and description, select an assignee where applicable, and optionally attach an image. The application validates the required fields and saves the request to Dataverse.

Each request receives a generated Request ID, such as BO-001, according to the configured auto-number settings.

4.4 My Requests

Employees and Managers can view requests they created or that were assigned to them. Available search and filtering controls help users locate requests by the supported fields and criteria.

4.5 Assigned Requests

Managers and Admins can access the Assigned Requests screen to review request information, open details, filter records, and perform the available status-management actions.

4.6 Request Details

The details screen displays the selected request's identifier, subject, type, priority, status, submission date, description, attached image if present, and available request updates.

4.7 Request Updates and Status Management

Users with the appropriate access can add request updates and perform the status actions available for a request. The supported status values are:

- Open
- In Progress
- Completed
- Canceled

Available actions depend on the request's current state and the configured application controls.

4.8 Image Management

Users can attach an optional image to a request, view the saved image on the details screen, and remove it through the implemented removal action. Removing an image clears its stored value from the corresponding Dataverse request record without deleting the request itself.

4.9 User Management

Admins can search demonstration user records and maintain configured user roles and account states.

4.10 Email Notifications

Power Automate processes assignment-related email notifications according to the configured workflow. The workflow is evaluated through its execution results, and email delivery depends on successful flow execution and the connected email service.

4.11 Full Dashboard

Managers and Admins can open the separate Power BI dashboard for more detailed operational reporting, subject to the configured report access and permissions.

5. Functional Requirements

The following requirements describe the core behavior of the tested application.

ID	Requirement	Priority
FR-01	The system shall validate demonstration login credentials against the configured user data in Dataverse.	High
FR-02	The system shall display navigation appropriate to the logged-in demonstration user's role.	High
FR-03	The system shall allow users to create requests using the required form fields and optional image attachment.	High
FR-04	The system shall validate required request fields before saving.	High
FR-05	The system shall save requests to Dataverse and generate a Request ID according to the configured auto-number settings.	High
FR-06	The system shall display requests created by or assigned to the current Employee or Manager, according to the configured filtering rules.	High
FR-07	The system shall allow Admin to view all requests.	High
FR-08	The system shall provide request search and filtering through the available controls.	High
FR-09	The system shall display the details and available update history of a selected request.	High
FR-10	The system shall support the configured request status actions and save successful changes.	High
FR-11	The system shall save and display optional request images using the configured Dataverse image column.	High
FR-12	The system shall clear the stored image value when the image-removal action is used.	High
FR-13	The system shall save request updates and display their available history.	High

FR-14	The system shall allow Admin to search users and update configured roles and account states.	High
FR-15	The configured Power Automate workflow shall process qualifying assignment-related email notifications.	High
FR-16	The application shall provide request summary information through the Power Apps dashboard.	High
FR-17	The Full Dashboard action shall open the configured Power BI report for authorized users.	High

6. Non-Functional Requirements

ID	Category	Requirement	Priority
NFR-01	Security	The application shall be identified as a demonstration system and shall not claim production-grade authentication.	Critical
NFR-02	Privacy	Only fictional identities and non-sensitive test data shall be used.	Critical
NFR-03	Authentication	Production deployment would require an approved authentication solution and appropriate credential management.	Critical
NFR-04	Authorization	Role-based navigation shall not replace appropriate data-level access controls.	Critical
NFR-05	Usability	Labels, navigation, validation messages, and notifications shall be clear and consistent.	High
NFR-06	Reliability	Success messages shall be displayed only after the relevant operation succeeds.	High
NFR-07	Data Integrity	Request identifiers shall be unique according to the Dataverse auto-number configuration.	High
NFR-08	Data Consistency	Request status changes and associated update history shall remain consistent when the corresponding operations succeed.	High
NFR-09	Attachment Integrity	Image removal shall clear the saved image value from the corresponding Dataverse record.	High
NFR-10	Performance	Lists and dashboards should load within an acceptable time for the demonstration dataset.	Medium
NFR-11	Maintainability	Important tables, fields, choice values, and application logic should be documented.	Medium

NFR-12	Compatibility	The application should be tested in the browser and screen sizes used for demonstration.	Medium
NFR-13	Availability	Cloud-based functionality requires network connectivity and availability of the relevant Microsoft services.	High
NFR-14	Auditability	Relevant workflow runs and important data-integrity test results should be available for review.	Medium

7. Technology Stack and Architecture

7.1 Technologies Used

Technology	Purpose
Microsoft Power Apps	Canvas application, user interface, navigation, forms, request lists, and internal dashboard.
Microsoft Dataverse	Primary data store for users, requests, request updates, and request images.
Microsoft Power Automate	Assignment-related email notification workflow.
Microsoft Power BI	Separate reporting dashboard and operational visualizations.

SQL, SharePoint, and Microsoft Entra ID are not part of the current connected application architecture. They may be considered for future development where appropriate.

7.2 High-Level Architecture

BankOps follows a low-code, cloud-based architecture.

1. **User Interface:** Users interact with the Power Apps Canvas application.
2. **Data Storage:** Power Apps reads and writes application data in Dataverse.
3. **Workflow Automation:** Power Automate processes qualifying assignment-related email notifications.
4. **Reporting:** Power BI provides a separate reporting dashboard for users with access.

7.3 Data Model

Entity	Purpose	Main Information
Users	Stores demonstration user profiles.	Name, email, role, department, employee ID, and account state.
Requests	Stores operational requests.	Request ID, subject, type, priority, description, status, submission date, image, and configured user lookups.
Request Updates	Stores notes and update history.	Related request, update text, status, author, and update date.

The exact fields and relationships follow the configured Dataverse schema.

7.4 Request Lifecycle

Current Status	Available Action	Result
Open	Start Processing	In Progress
Open	Cancel Request	Canceled
In Progress	Complete Request	Completed
In Progress	Cancel Request	Canceled
Completed	No further lifecycle action in the current interface	Completed
Canceled	No further lifecycle action in the current interface	Canceled

The available actions depend on the implemented controls and authorization rules.

7.5 Image Storage and Removal

Request images are stored using the configured Dataverse image column. The application can display the saved image on Request Details and remove it through the implemented removal action. The removal operation clears the image value while preserving the request record and its other data.

8. External Interfaces

8.1 User Interface

BankOps is accessed through Power Apps in a supported web browser. Its main screens include Home, New Request, My Requests, Assigned Requests, Request Details, Full Dashboard, and User Management, according to the user's role.

8.2 Dataverse Interface

Dataverse is the application's primary data source. Power Apps uses it to retrieve and save user records, requests, request updates, and image data.

8.3 Email Interface

Power Automate connects the application workflow to the configured email service. The flow processes qualifying notifications, and its run history can be used to review execution results.

8.4 Reporting Interface

The Full Dashboard function opens the configured Power BI report. Access depends on the report configuration, sharing permissions, licensing, and data-source access.

8.5 Operating Requirements

The application requires a supported browser, network connectivity, and access to the relevant Microsoft cloud services. Offline operation is not included in the current design.

9. Use Case Overview

9.1 Actors

Actor	Main Activities
Employee	Log in, create requests, view My Requests, view request details, and track progress.
Manager	Perform Employee functions, access Assigned Requests, manage permitted request statuses, and open Full Dashboard.
Admin	Perform the standard request functions, view all requests, access Assigned Requests and Full Dashboard, and manage users.

The email service is an external system integration, not a fourth human actor.

9.2 Main Use Cases

Use Case	Description	Primary Actor
UC-01 — Log In	Validate demonstration credentials and open the appropriate role-based interface.	Employee, Manager, Admin
UC-02 — Create Request	Enter request information, optionally attach an image, and save the request	Employee and other roles where enabled
UC-03 — View Requests	Search, filter, and open requests within the user's configured scope.	Employee, Manager, Admin
UC-04 — Manage Request	Review request details and perform permitted status actions.	Manager, Admin, or other authorized users
UC-05 — Add Request Update	Save a note associated with a request and view its history.	Authorized user
UC-06 — Remove Request Image	Clear the saved image from the selected Dataverse request record.	Authorized user
UC-07 — Manage Users	Search user records and update configured roles or account states.	Admin
UC-08 — Send Email Notification	Process a qualifying assignment notification through Power Automate.	Automated workflow
UC-09 — View Full Dashboard	Open the Power BI reporting dashboard.	Manager, Admin

10. Verification and Demonstration

The BankOps application was tested using the Employee, Manager, and Admin roles. Testing covered the main application functions, including request management, role-based access, image handling, workflow automation, and reporting.

Test Area	Verification
Role-Based Access	Verified the application navigation and request visibility for the configured roles.
Request Management	Tested request creation, request details, status changes, and request updates.
Image Management	Verified image display and confirmed that image removal clears the stored image value in Dataverse.

User Management	Tested user search and the configured role and account-state updates.
Email Notifications	Checked the Power Automate workflow execution and the corresponding email result.
Reporting	Tested access to the Power Apps dashboard and the Power BI reporting dashboard.

Appendix A — Application Screenshots

This appendix presents selected screenshots of the BankOps application to demonstrate its main interfaces and implemented features.

Figure A.1 — Login Screen

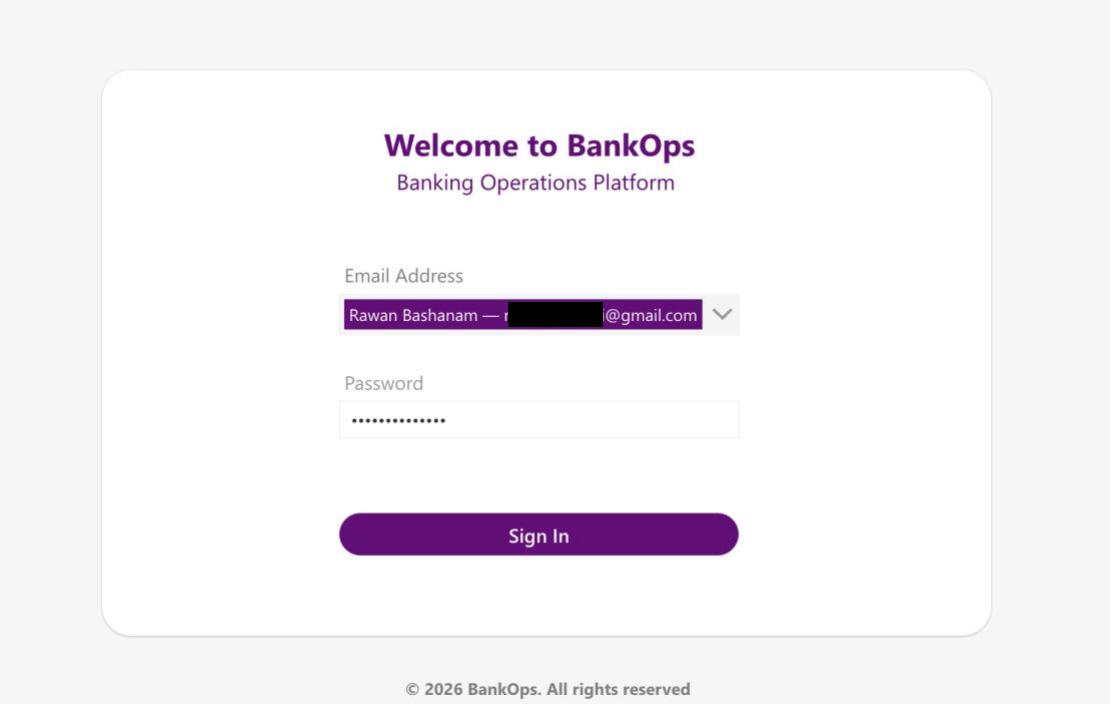


Figure A.2 — Employee Home Screen

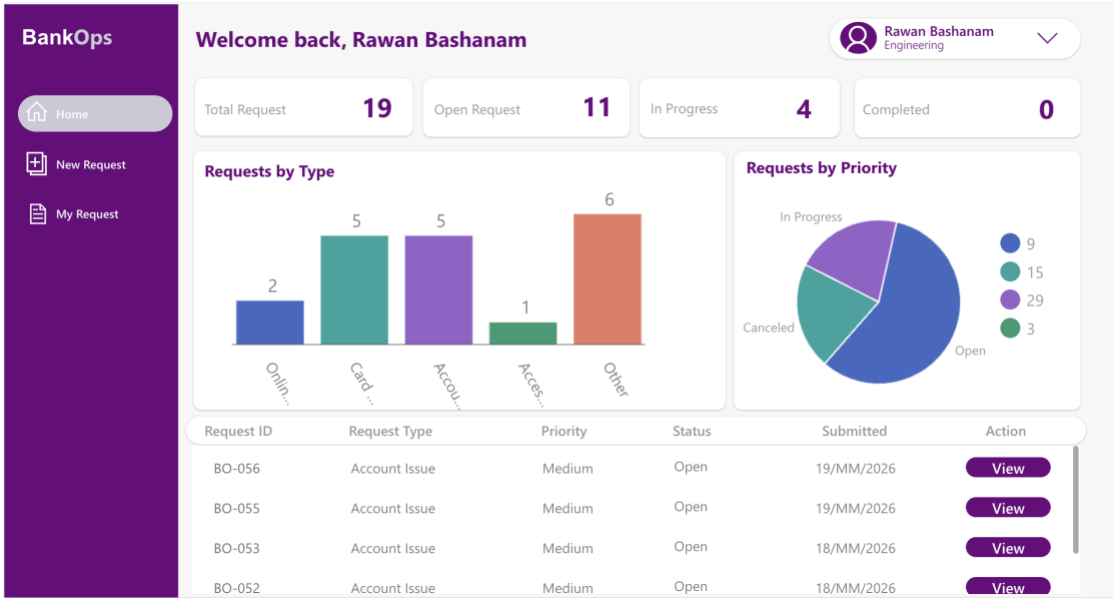


Figure A.3 — New Request Screen

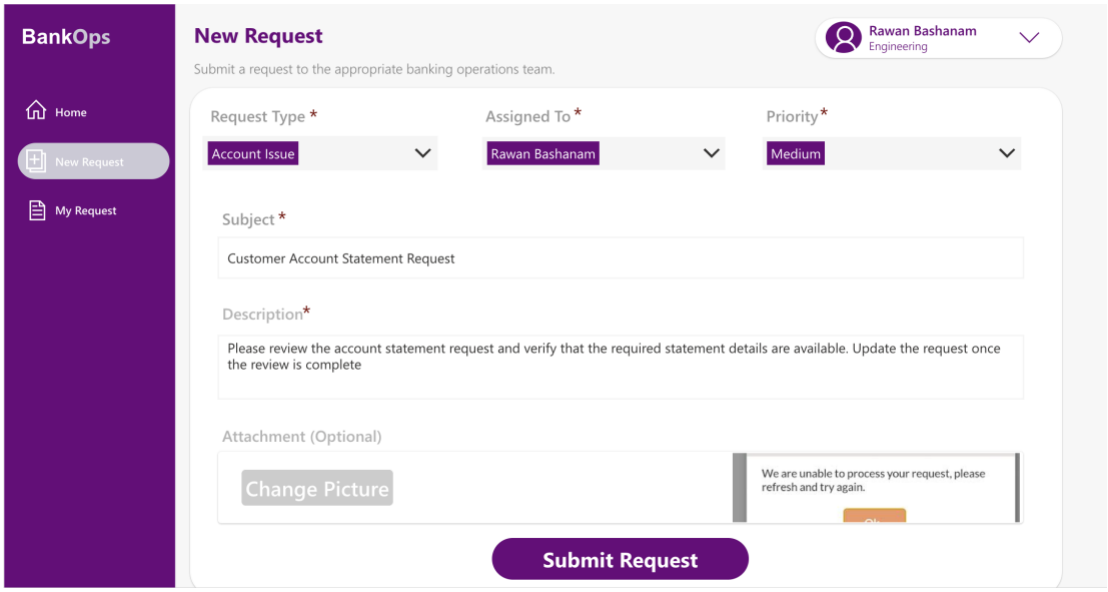


Figure A.4 — My Requests Screen

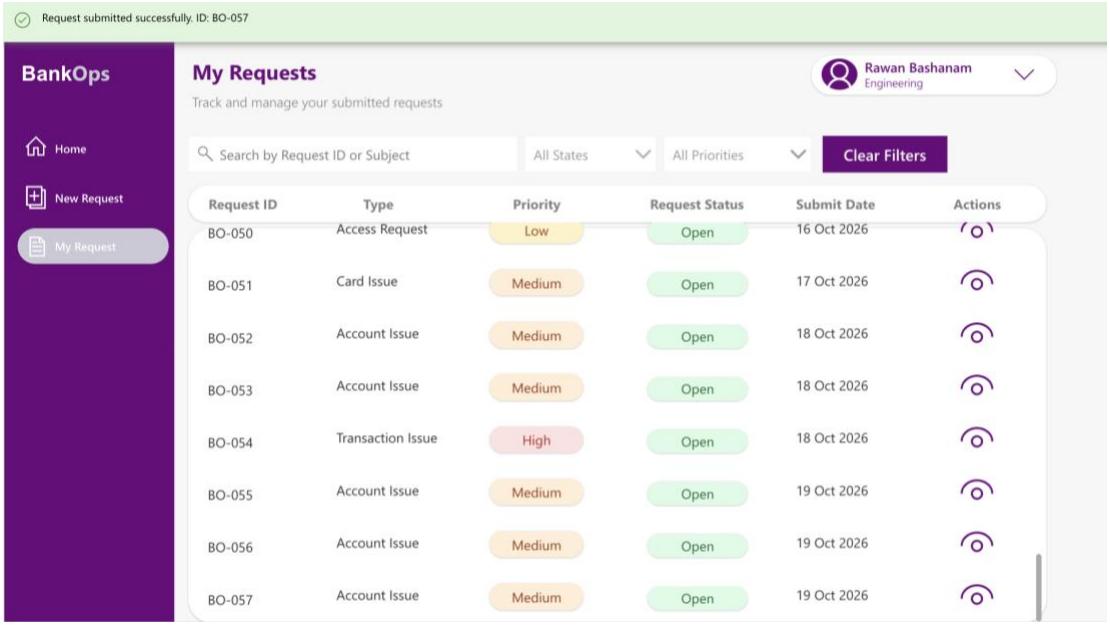
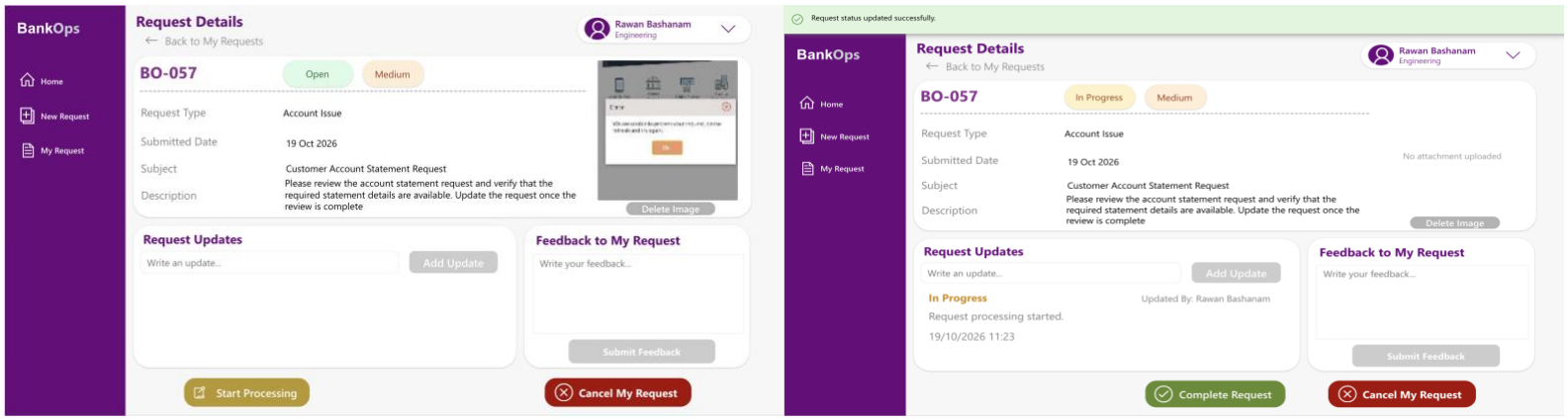


Figure A.5 — Request Details Screen



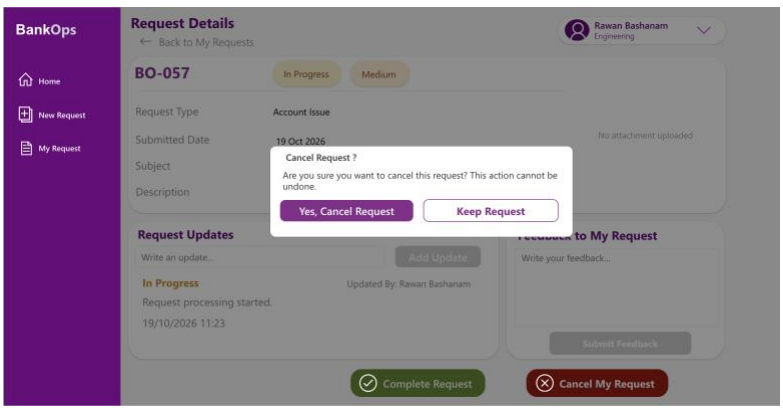


Figure A.6 — Request Details After Image Removal

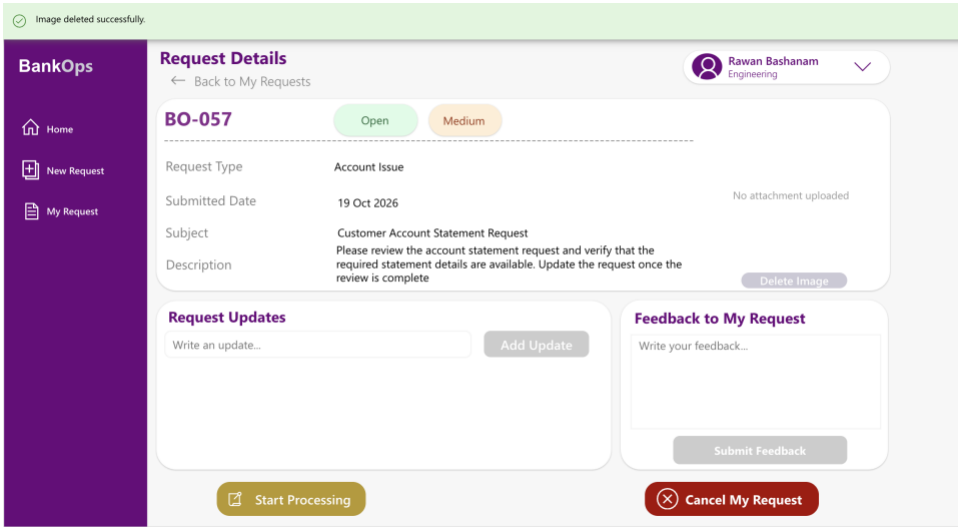


Figure A.7 — Manager Assigned Requests Screen

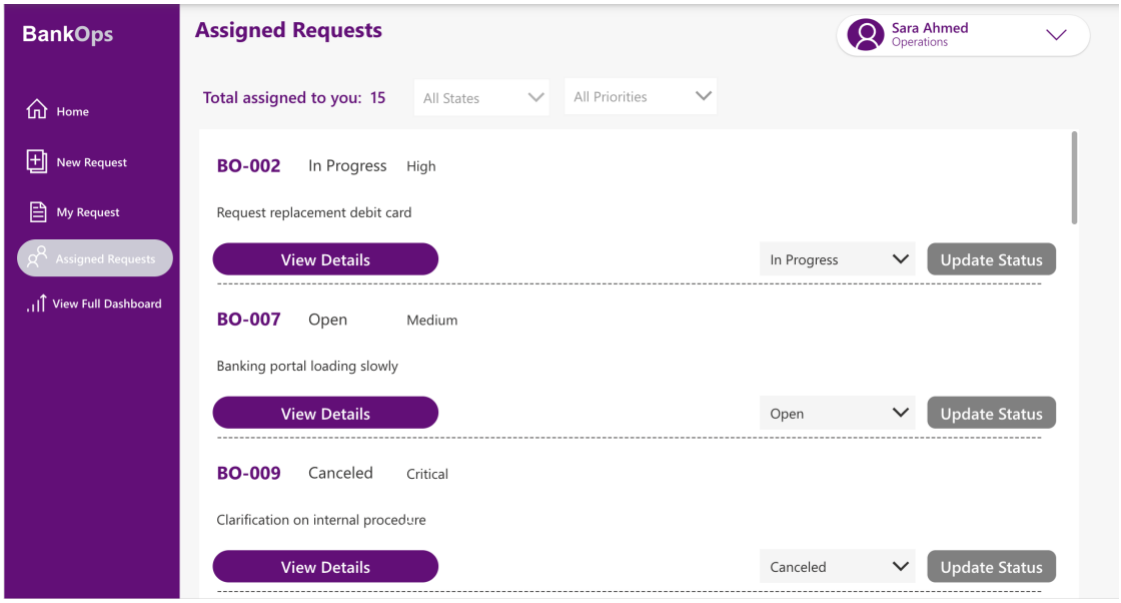


Figure A.8 — Admin User Management Screen

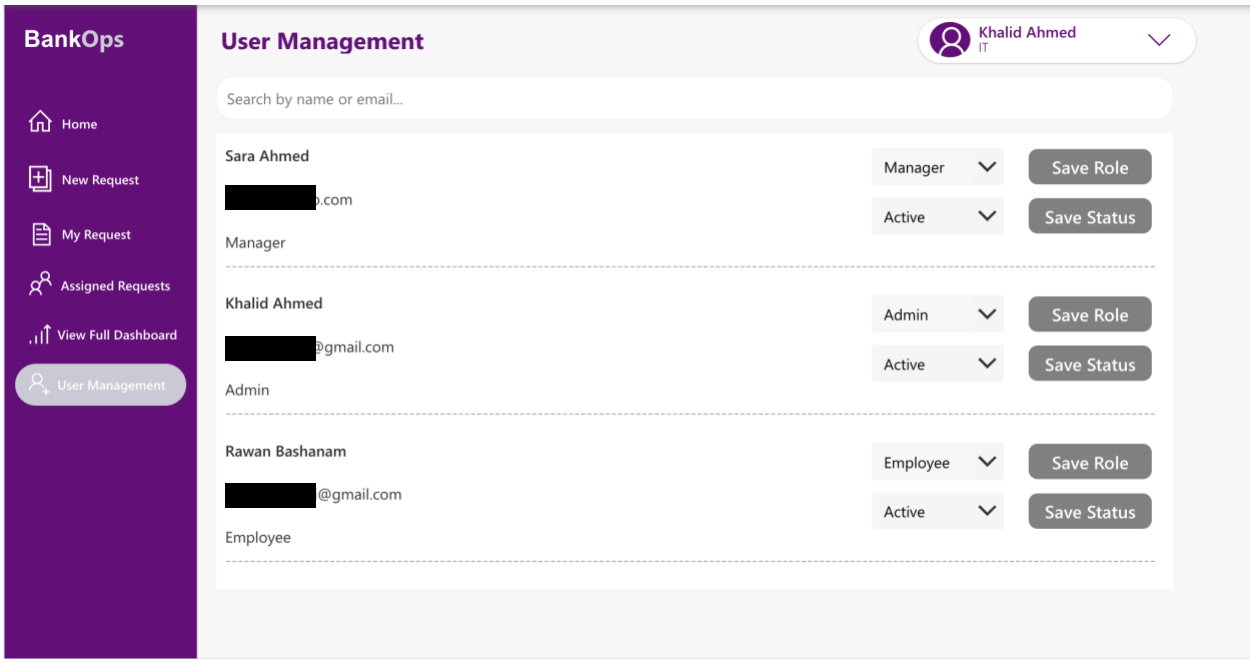


Figure A.9 — Power Automate Assignment Email Notification

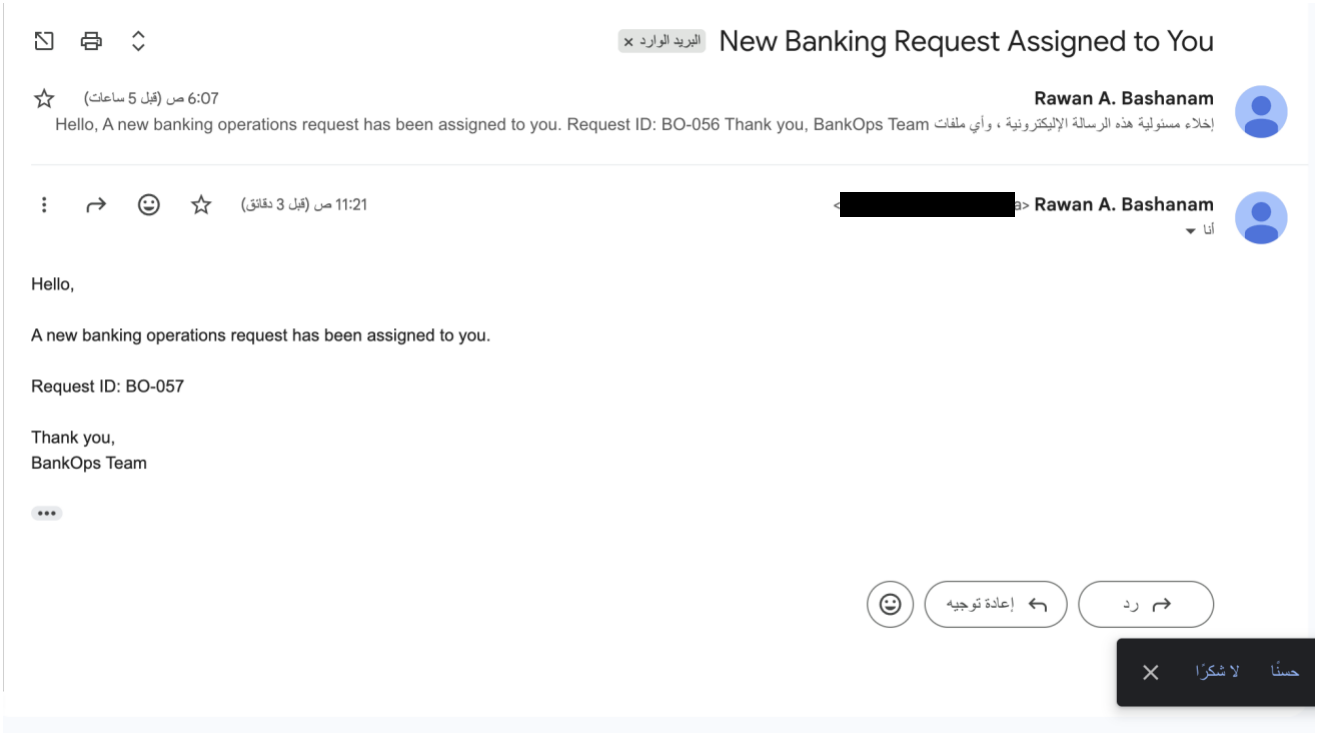
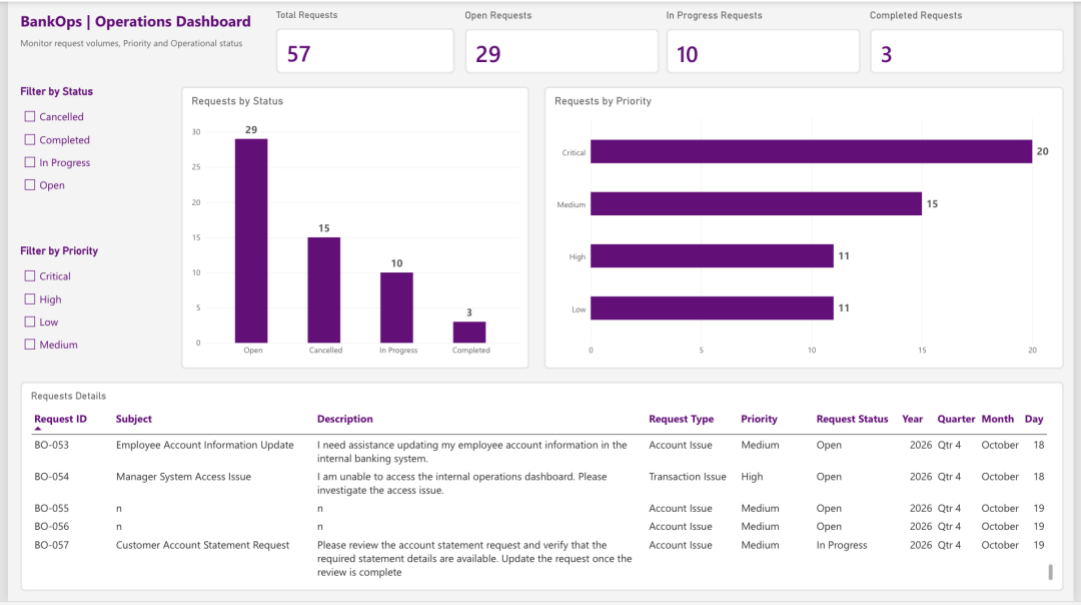


Figure A.10 — Power BI Reporting Dashboard



Appendix B — Project Limitations and Future Improvements

B.1 Project Limitations

BankOps is a functional portfolio project developed in a simulated banking environment. It uses fictional users and test data and is not intended for production banking operations.

The current application uses a demonstration login mechanism based on user data stored in Dataverse. It does not use Microsoft Entra ID authentication or provide production-grade credential security.

SQL and SharePoint are not connected to the current application. The application also depends on network connectivity and the availability of the relevant Microsoft cloud services.

B.2 Future Improvements

Potential future improvements include:

- Integrating Microsoft Entra ID or another approved identity provider for authentication.
- Reviewing and strengthening data access permissions and security controls.
- Expanding Power Automate workflows and notification monitoring.
- Adding further reporting and operational analytics.
- Evaluating SQL or SharePoint integration if future business requirements justify it.